



sors have been set up in such a way that one is upwind from the city and the second downwind, covering as many wind directions as possible. With the input of the sensor data and meteorological parameters, high-performance computers can create a spatially resolved emission map of the city.

Prof. Jia Chen and project manager Florian Dietrich at one of the sensors for measuring urban greenhouse gas emissions (Credit: www.tum.de/nc/en)

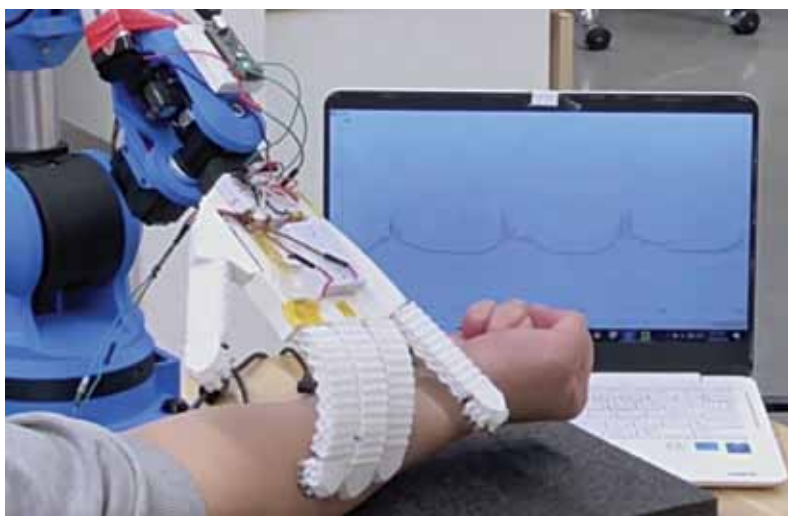
AI and wearable cameras for self-walking robotic exoskeletons

Robotics researchers at the University of Waterloo are developing exoskeleton legs capable of thinking and making control decisions on their own using sophisticated artificial intelligence (AI) technology. The system combines computer vision and deep-learning AI to mimic how able-bodied people walk by seeing their surroundings and adjusting their movements. To address the limitation of manually controlling the motors, the researchers fitted exoskeleton users with wearable cameras and are now optimising AI computer software to process the video feed to accurately recognise stairs, doors, and other features of the surrounding environment. The researchers are also working to improve the energy efficiency of motors for robotic exoskeletons by using human motion to self-charge the batteries.



Use of AI and wearable cameras (Credit: <https://uwaterloo.ca/news>)

Soon to be rolled out robot healthcare helpers



Sensing robot healthcare helpers (Credit: www.sfu.ca/university-communications)

Sensing robots that could perform healthcare tasks for supporting doctors and nurses are being developed by researchers at the School of Mechatronic Systems Engineering, Simon Fraser University. Created using 3D printed structures, the robot contains biomedical electrodes at the tip of each finger, which allows detection and measurement of human physiological signals on touching a person. These signals include electrocardiogram (which monitors heartbeat), respiration rate, electromyogram (monitoring electrical signals from muscle movements), and temperature. The humanoid robot can also monitor oxygen levels, beneficial for supervising the condition of those who develop severe Covid-19. The data can be viewed in real time on the robot's monitor or sent directly to the healthcare provider.